607. A Memoir on Prepotentials (continued)

[Continuation of the same paper; articles 60–75 of the Annexes.]

[p. 356] and the first term of this being $= \sqrt{e^2+R^2} - e$, this is consequently

$$= \sqrt{R^2+e^2} + \tfrac{1}{2}e \int_0^X x^{-\frac{1}{2}} \left[1 - (1-x)^{\frac{1}{2}s-1}\right] dx - e\left(1 + \tfrac{1}{2}\int_0^1 x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx\right).$$

As regards the second term of this, we have

$$-2x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right] + 2\left(\tfrac{1}{2}s-1\right)\int x^{-\frac{1}{2}}(1-x)^{\frac{1}{2}s-2}\,dx = \int x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx;$$

or, taking each term between the limits 1, 0,

$$-2 + 2\left(\tfrac{1}{2}s-1\right)\frac{\Gamma\frac{1}{2}\,\Gamma(\frac{1}{2}s-1)}{\Gamma(\frac{1}{2}s-\frac{1}{2})} = \int_0^1 x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx;$$

viz. this integral has the value

$$-2 + \frac{2\,\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

and the value of the r-integral $\displaystyle\int_0^R \frac{r^{s-1}\,dr}{(r^2+e^2)^{\frac{1}{2}s+q}}$ is consequently

$$= \sqrt{R^2+e^2} + \tfrac{1}{2}e\int_0^X x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx - e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

which is of the form

$$R\left\{1 + \text{terms in } \frac{e^2}{R^2}, \frac{e^4}{R^4}, \ldots\right\} - e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})};$$

say the approximate value is

$$R - e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

where the first term R is the term $\displaystyle\int_0^R dr$, given by the expansion in ascending powers of e^2; the second term is the term in e^{-1}. And observe that the term is the value of

$$\tfrac{1}{2}e\int_0^1 x^{-\frac{1}{2}}(1-x)^{\frac{1}{2}s-1}\,dx,$$

calculated by means of the ordinary formula for a Eulerian integral (which formula, on account of the negative exponent $-\frac{1}{2}$, is not really applicable, the value of the integral being $=\infty$) on the assumption that the Γ of a negative q is interpreted in accordance with the equation $\Gamma(q+1) = q\Gamma q$; viz. the value thus calculated is

$$\tfrac{1}{2}e\,\frac{\Gamma(-\frac{1}{2})\,\Gamma(\frac{1}{2}s)}{\Gamma(\frac{1}{2}s-\frac{1}{2})} = -e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

on the assumption $\Gamma(-\frac{1}{2}) = -\frac{1}{2}\,\Gamma(-\frac{1}{2})$ *[scan reads $-\frac{1}{2}\Gamma(-\frac{1}{2})$ for the recursion]*; and this agrees with the foregoing value.

60. It is now easy to see in general how the foregoing transformed value $\frac{1}{2}e^{-q}\displaystyle\int_X^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx$, where q is negative and fractional, gives at once the value of

[p. 357] the term in e^{-q}. Observe that in the integral x is always between 1 and X $\left(=\dfrac{e^2}{e^2+R^2}\right)$, a positive quantity less than 1; the function to be integrated never becomes infinite. Imagine for a moment an integral $\int_X^\infty x^a\,dx$, where a is positive or negative. We may conventionally write this $= \int_0^\infty x^a\,dx - \int_0^X x^a\,dx$, understanding the first symbol to mean $\dfrac{1^{1+a}}{1+a}$, and the second to mean $\dfrac{X^{1+a}}{1+a}$; they of course properly mean $\dfrac{1^{1+a}-0^{1+a}}{1+a}$ and $\dfrac{X^{1+a}-0^{1+a}}{1+a}$; but the terms in 0^{1+a}, whether zero or infinite, destroy each other, the original form $\int_X^\infty x^a\,dx$, in fact, showing that no such terms can appear in the result.

In accordance with the convention, we write

$$\int_X^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx = \int_0^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx - \int_0^X x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx;$$

and it follows that the term in e^{-q} is

$$\tfrac{1}{2}e^{-q}\int_0^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx,$$

this last expression (wherein q, it will be remembered, is a negative fraction) being understood according to the convention; and so understanding it, the value of the term is

$$= \tfrac{1}{2}e^{-q}\,\frac{\Gamma q\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)},$$

where the Γ of the negative q is to be interpreted in accordance with the equation $\Gamma(q+1) = q\Gamma q$; viz. we have $\Gamma q = \dfrac{1}{q}\,\Gamma(q+1),\ = \dfrac{1}{q(q+1)}\,\Gamma(q+2)$, &c., so as to make the argument of the Γ positive. Observe that under this convention we have

$$\Gamma q\,\Gamma(1-q) = \frac{\Gamma(\frac{1}{2})^2}{\sin q\pi} = \frac{\pi}{\sin q\pi};\quad \text{or the term is}\quad \tfrac{1}{2}e^{-q}\,\frac{\pi}{\sin q\pi}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1-q)}.$$

61. An example in which $-\frac{1}{2}s-q-1$ is integral will make the process clearer, and will serve instead of a general proof. Suppose $q=-\frac{7}{2}$, $\frac{1}{2}s=\frac{1}{4}$, the expression

$$\int_0^1 x^{-\frac{9}{2}}(1-x)^{\frac{1}{4}}\,dx = \int_0^1 \left(x^{-\frac{9}{2}} - \tfrac{9}{2}\,x^{-\frac{7}{2}} + \tfrac{63}{8}\,x^{-\frac{5}{2}} - \tfrac{63}{16}\,x^{-\frac{3}{2}} + \cdots\right)dx$$

is used, in accordance with the convention, to denote the value

$$= -\tfrac{2}{7} + \tfrac{2}{5}\cdot\tfrac{9}{2} - \tfrac{2}{3}\cdot\tfrac{63}{8} + \tfrac{2}{1}\cdot\tfrac{63}{16} = 7\left(-\tfrac{2}{7\cdot1} + \tfrac{2}{5\cdot3} - \tfrac{2}{3\cdot5} + \tfrac{2}{1\cdot7}\right) = \frac{-7\cdot2401}{5\cdot13\cdot27} = \frac{-7^4}{5\cdot13\cdot27}.$$

[p. 358] But we have

$$\frac{\Gamma\frac{1}{2}s\,\Gamma q}{\Gamma(\frac{1}{2}s+q)} = \frac{\Gamma\frac{5}{4}\,\Gamma(-\frac{7}{2})}{\Gamma(\frac{5}{4}-\frac{7}{2})} = \frac{2^4\,\Gamma(-\frac{7}{2})}{\frac{1}{4}\cdot\frac{5}{4}\cdot\frac{9}{4}\cdot\frac{13}{4}\cdot\Gamma(-\frac{7}{2})} = \frac{2^4\cdot4^4}{5\cdot13\cdot27} = \frac{-7^4}{5\cdot13\cdot27},$$

agreeing with the former value.

62. The case of a negative integer is more simple. To find the logarithmic term of

$$\tfrac{1}{2}e^{-q}\int_X^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx,$$

we have only to expand the factor $(1-x)^{\frac{1}{2}s-1}$ so as to obtain the term involving x^{-q}.

We have thus the term

$$\tfrac{1}{2}e^{-q}\,(-)^{-q}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(1-q)\,\Gamma(\frac{1}{2}s+q)}\,e^{-q}\log\tfrac{1}{X},$$

where $\log\frac{1}{X} = \log\left(1+\dfrac{R^2}{e^2}\right) = 2\log\dfrac{R}{e} + 2\log\sqrt{1+\dfrac{e^2}{R^2}}$; so that, neglecting the terms in $\dfrac{e^2}{R^2}$, &c., this is $2\log\dfrac{R}{e}$, and the term in question is

$$= (-)^q e^{-q}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(1-q)\,\Gamma(\frac{1}{2}s+q)}\ \log\frac{R}{e}.$$

The general conclusion is that q being negative, the r-integral

$$\int_0^R \frac{r^{s-1}\,dr}{(r^2+e^2)^{\frac{1}{2}s+q}}$$

has for its value a series proceeding in powers of e^2, which series up to a certain point is equal to the series obtained by expanding in ascending powers of e^2 and integrating each term separately; viz. the series to the point in question is

$$\frac{R^{-2q}}{-2q} - \frac{\frac{1}{2}s+q}{1}\cdot\frac{R^{-2q-2}}{-2q-2}\,e^2 + \frac{\frac{1}{2}s+q\cdot\frac{1}{2}s+q+1}{1\cdot 2}\cdot\frac{R^{-2q-4}}{-2q-4}\,e^4 - \cdots,$$

continued so long as the exponent of e is less than $-2q$: together with a term Ke^{-q} when q is fractional, and $Ke^{-q}\log\dfrac{R}{e}$ when q is integral; viz. q fractional, this term is

$$= \tfrac{1}{2}e^{-q}\,\frac{\Gamma q\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)} = \tfrac{1}{2}e^{-q}\,\frac{\pi}{\sin q\pi}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1-q)},$$

and q integral, it is

$$= (-)^q e^{-q}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(1-q)\,\Gamma(\frac{1}{2}s+q)}\ \log\frac{R}{e}.$$

[p. 359] 63. It has been tacitly assumed that $\frac{1}{2}s+q$ is positive; but the formulae hold good if $\frac{1}{2}s+q$ is $=0$ or negative. Suppose $\frac{1}{2}s+q$ is $=0$ or a negative integer, then $\Gamma(\frac{1}{2}s+q)=\infty$, and the special term involving e^{-q} or $e^{-q}\log e$ vanishes; in fact, in this case the r-integral is

$$= \int_0^R r^{s-1}\,(r^2+e^2)^{-(\frac{1}{2}s+q)}\,dr,$$

where $(r^2+e^2)^{-(\frac{1}{2}s+q)}$ has for its value a finite series, and the integral is therefore equal to a finite series $A+Be^2+Ce^4+$&c. If $\frac{1}{2}s+q$ be fractional, then the Γ of the negative quantity $\frac{1}{2}s+q$ must be understood as above, or, what is the same thing, we may, instead of $\Gamma(\frac{1}{2}s+q)$, write

$$\frac{\pi}{\sin(\frac{1}{2}s+q)\pi\cdot\Gamma(1-q-\frac{1}{2}s)};$$

thus, q being integral, the exceptional term is

$$= \frac{(-)^q\sin(\frac{1}{2}s+q)\pi\cdot\Gamma(1-q-\frac{1}{2}s)}{\Gamma(\frac{1}{2}s)\,\Gamma(1-q)}\,\Gamma\tfrac{1}{2}s\cdot R^{-2q}e^{-q}.$$

For instance, $s=1$, $q=-2$, the term is

$$\tfrac{1}{2}\,\frac{\Gamma\frac{1}{2}\sin(-\frac{3}{2})\pi\cdot\Gamma\frac{5}{2}}{\Gamma\frac{1}{2}\cdot\Gamma 3}\,e^2\log\frac{R}{e},$$

or, since $\Gamma\frac{5}{2} = \frac{3}{2}\cdot\frac{1}{2}\,\Gamma\frac{1}{2}$, and $\Gamma 3 = 2$, the term is $+\frac{3}{8}e^2\log\dfrac{R}{e}$, agreeing with a preceding result.

Annex III. Prepotentials of Uniform Spherical Shell and Solid Sphere.
Art. Nos. 64 to 92.

64. The prepotentials in question depend ultimately upon two integrals, which also arise, as will presently appear, from prepotential problems in two-dimensional space, and which are for convenience termed the *ring-integral* and the *disk-integral* respectively. The analytical investigation in regard to these, depending as it does on a transformation of a function allied with the hypergeometric series, is I think interesting.

65. Consider first the prepotential of a uniform $(s+1)$-dimensional spherical shell. This is

$$V = \int \frac{dS}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}},$$

the equation of the surface being $x^2 + \ldots + z^2 + w^2 = f^2$; and there are the two cases of an internal point, $a^2 + \ldots + c^2 + e^2 < f^2$, and an external point, $a^2 + \ldots + c^2 + e^2 > f^2$.

The value is a function of $a^2 + \ldots + c^2 + e^2$, say this is $= \kappa^2$. Taking the axes so that the coordinates of the attracted point are $(0, \ldots, 0, \kappa)$, the integral is

$$= \int \frac{dS}{\{x^2 + \ldots + z^2 + (\kappa - w)^2\}^{\frac{1}{2}s+q}},$$

[p. 360] where the equation of the surface is still $x^2 + \ldots + z^2 + w^2 = f^2$. Writing $x = f\xi, \ldots, z = f\zeta$, $w = f\omega$, where $\xi^2 + \ldots + \zeta^2 + \omega^2 = 1$, we have $dS = f^s \dfrac{d\xi \ldots d\zeta}{\omega}$, or the integral is

$$= f^s \int \frac{d\xi \ldots d\zeta}{\omega\,(f^2 - 2\kappa f\omega + \kappa^2)^{\frac{1}{2}s+q}}.$$

Assume $\xi = \rho x, \ldots, \zeta = \rho z$, where $x^2 + \ldots + z^2 = 1$; then $\rho^2 + \omega^2 = 1$. Moreover, $d\xi \ldots d\zeta = \rho^{s-1}\,d\rho\,d\Sigma$, where $d\Sigma$ is the element of surface of the s-dimensional unit-sphere $x^2 + \ldots + z^2 = 1$; or for ρ, substituting its value $\sqrt{1-\omega^2}$, we have $d\rho = -\dfrac{\omega\,d\omega}{\sqrt{1-\omega^2}}$; and thence $d\xi \ldots d\zeta = -(1-\omega^2)^{\frac{1}{2}s-\frac{3}{2}}\,\omega\,d\omega\,d\Sigma$. The integral as regards ρ is from $\rho = -1$ to $+1$, or as regards ω from 1 to -1; whence reversing the sign, the integral will be from $\omega = -1$ to $+1$; and the required integral is thus

$$= f^s \int d\Sigma \int_{-1}^{+1} \frac{(1-\omega^2)^{\frac{1}{2}s-\frac{3}{2}}\,d\omega\,d\Sigma}{(f^2 - 2\kappa f\omega + \kappa^2)^{\frac{1}{2}s+q}} = f^s \int d\Sigma \int_{-1}^{+1} \frac{(1-\omega^2)^{\frac{1}{2}s-\frac{3}{2}}\,d\omega}{(f^2 - 2\kappa f\omega + \kappa^2)^{\frac{1}{2}s+q}},$$

where $\displaystyle\int d\Sigma$ is the surface of the s-dimensional unit-sphere (see Annex I.), $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$; and for greater convenience transforming the second factor by writing therein $\omega = \cos\theta$, the required integral is $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\,f^s$ multiplied by

$$\int_0^\pi \frac{\sin^{s-1}\theta\,d\theta}{(f^2 - 2\kappa f\cos\theta + \kappa^2)^{\frac{1}{2}s+q}},$$

which last expression including the factor $2f^s$, but without the factor $\dfrac{(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$, is the ring-integral discussed in the present Annex. It may be remarked that the value can be at once obtained in the particular case $s = 2$, which belongs to tridimensional space; viz. we then have

$$V = \int_0^\pi \frac{2\pi f^2 \sin\theta\,d\theta}{(f^2 - 2f\kappa\cos\theta + \kappa^2)^{q+1}} = \frac{2\pi f}{\kappa(1-q)}\big|\,(f^2 - 2f\kappa\cos\theta + \kappa^2)^{-q+1}\big|,$$

which agrees with a result given, *Mécanique Céleste*, Book XII. Chap. II.

66. Consider next the prepotential of the uniform solid $(s+1)$-dimensional sphere,

$$V = \int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}},$$

the equation of the surface being $x^2 + \ldots + z^2 + w^2 = f^2$; there are the two cases of an internal point $\kappa < f$, and an external point $\kappa > f$ $(a^2 + \ldots + c^2 + e^2 = \kappa^2$ as before).

[p. 361] Transforming so that the coordinates of the attracted point are $0, \ldots, 0, \kappa$, the integral is

$$= \int \frac{dx \ldots dz\, dw}{\{x^2 + \ldots + z^2 + (\kappa - w)^2\}^{\frac{1}{2}s+q}},$$

where the equation is still $x^2 + \ldots + z^2 + w^2 = f^2$. Writing here $x = r\xi, \ldots, z = r\zeta$, where $\xi^2 + \ldots + \zeta^2 = 1$, we have $dx \ldots dz = r^{s-1}\, dr\, dS$, where dS is an element of surface of the s-dimensional unit-sphere $\xi^2 + \ldots + \zeta^2 = 1$; the integral is therefore

$$= \int\int \frac{r^{s-1}\, dr\, dS\, dw}{\{r^2 + (\kappa - w)^2\}^{\frac{1}{2}s+q}},$$

where, as regards r and w, the integration extends over the circle $r^2 + w^2 = f^2$. The value of the first factor (see Annex I.) is $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$; writing y and x in place of r and w respectively, the integral is $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$ multiplied by

$$\int \frac{y^{s-1}\, dx\, dy}{\{(x-\kappa)^2 + y^2\}^{\frac{1}{2}s+q}}$$

over the circle $x^2 + y^2 = f^2$; viz. this last expression (without the factor $\dfrac{(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$) is the disk-integral discussed in the present Annex.

67. We find, for the value in regard to an internal point $\kappa < f$,

$$V = \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, f^{s+1} \int_0^\infty (t + f^2 - \kappa^2)^{q-\frac{1}{2}}\, t^{-q-1}\, (t+f^2)^{\frac{1}{2}s+q-\frac{1}{2}}\, dt,$$

which, in the particular case $q = -\frac{1}{2}$, is

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})}\, f^{s+1} \int_0^\infty (t + f^2 - \kappa^2)^{-1}\, (t+f^2)^{\frac{1}{2}s-1}\, dt;$$

viz. the integral in t is here

$$= \int_{f^2-\kappa^2}^\infty (t + f^2 - \kappa^2)^{-1}\, t^{-\frac{1}{2}}\, (t+f^2)^{\frac{1}{2}s-1}\, dt = \frac{1}{f^{s+1}} \left(\frac{f^{s+1}}{\frac{1}{2}s-\frac{1}{2}} - \frac{\kappa^{s+1}}{\frac{1}{2}s+\frac{1}{2}} \right),$$

or we have

$$V = \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})} \left(\frac{f^{s+1}}{\frac{1}{2}s-\frac{1}{2}} - \frac{\kappa^{s+1}}{\frac{1}{2}s+\frac{1}{2}} \right).$$

It may be added that, in regard to an external point $\kappa > f$, the value is

$$V = \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, f^{s+1} \int_{\kappa^2-f^2}^\infty (t + f^2 - \kappa^2)^{q-\frac{1}{2}}\, t^{-q-1}\, (t+f^2)^{\frac{1}{2}s+q-\frac{1}{2}}\, dt,$$

C. IX. 46

[p. 362] which, in the same case $q = -\frac{1}{2}$, is

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})}\, f^{s+1} \int_{\kappa^2-f^2}^\infty (t + f^2 - \kappa^2)^{-1}\, (t+f^2)^{\frac{1}{2}s-1}\, dt,$$

where the t-integral is

$$= \int_{\kappa^2-f^2}^{\infty} (t+f^2-\kappa^2)^{-1}\, t^{-\frac{1}{2}}\, (t+f^2)^{\frac{1}{2}s-1}\, dt = \frac{\kappa^{s+1}}{\frac{1}{2}s-\frac{1}{2}} \cdot \frac{\kappa^{s-2}}{\frac{1}{2}s+\frac{1}{2}} \cdot \frac{1}{\kappa^{s+1}} = \frac{\kappa^{s-1}}{\frac{1}{2}s-\frac{1}{2}\cdot\frac{1}{2}s+\frac{1}{2}};$$

and the value of V is therefore

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\, \frac{f^{s+1}}{\kappa^{s-1}}.$$

Recurring to the case of the internal point; then, writing $\nabla = \dfrac{d^2}{da^2} + \ldots + \dfrac{d^2}{dc^2} + \dfrac{d^2}{de^2}$, and observing that $\nabla\,(\kappa^s) = s\,(\frac{1}{2}s+\frac{1}{2})$, we have

$$\nabla V = -\frac{4\,(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})};$$

(in particular, for ordinary space $s+1=3$, or the value is $\dfrac{-4\pi^2}{\sqrt{\pi}} = -4\pi$, which is right).

68. The integrals referred to as the ring-integral and the disk-integral arise also from the following integrals in two-dimensional space, viz. these are

$$\int \frac{y^{s-1}\, dS}{\{(x-\kappa)^2+y^2\}^{\frac{1}{2}s+q}}, \qquad \int \frac{y^{s-1}\, dx\, dy}{\{(x-\kappa)^2+y^2\}^{\frac{1}{2}s+q}},$$

in the first of which dS denotes an element of arc of the circle $x^2+y^2=f^2$, the integration being extended over the whole circumference, and in the second the integration extends over the circle $x^2+y^2=f^2$; y^{s-1} is written for shortness instead of $(y^2)^{\frac{1}{2}(s-1)}$, viz. this is considered as always positive, whether y is positive or negative; it is moreover assumed that $s-1$ is zero or positive.

Writing in the first integral $x = f\cos\theta$, $y = f\sin\theta$, the value is

$$= f^s \int \frac{(\sin\theta)^{s-1}\, d\theta}{(f^2-2\kappa f\cos\theta+\kappa^2)^{\frac{1}{2}s+q}},$$

viz. this represents the prepotential of the circumference of the circle, density varying as $(\sin\theta)^{s-1}$, in regard to a point $x=\kappa$, $y=0$ in the plane of the circle; and similarly the second integral represents the prepotential of the circular disk, density of the element at the point $(x,y) = y^{s-1}$, in regard to the same point $x=\kappa$, $y=0$; it being in each case assumed that the prepotential of an element of mass $\rho\, d\varpi$ at a point at distance d is $= \dfrac{\rho\, d\varpi}{d^{2q+s}}$.

[p. 363] 69. In the case of the circumference, it is assumed that the attracted point is not on the circumference, κ not $= f$; and the function under the integral sign, and therefore the integral itself, is in every case finite. In the case of the circle, if κ be an interior point, then if $2q-1$ be $=0$ or positive, the element at the attracted point becomes infinite; but to avoid this we consider, not the potential of the whole circle, but the potential of the circle less an indefinitely small circle radius ϵ having the attracted point for its centre; which being so, the element under the integral sign, and consequently the integral itself, remains finite.

It is to be remarked that the two integrals are connected with each other; viz. the circle of the second integral being divided into rings by means of a system of circles concentric with the bounding circle $x^2+y^2=f^2$, then the prepotential of each ring or annulus is determined by an integral such as the first integral; or, analytically, writing in the second integral $x = r\cos\theta$, $y = r\sin\theta$, and therefore $dx\,dy = r\,dr\,d\theta$, the second integral is

$$= \int r^s\, dr \int \frac{(\sin\theta)^{s-1}\, d\theta}{(r^2+\kappa^2-2fr\cos\theta)^{\frac{1}{2}s+q}},$$

viz. the integral in regard to θ is here the same function of r, κ that the first integral is of f, κ; and the integration in regard to r is of course to be taken from $r=0$ to $r=f$. But the θ-integral is not, in its original form, such a function of r as to render possible the integration in regard to r; and I, in fact, obtain the second integral by a different and in some respects a better process.

70. Consider first the ring-integral which, writing therein as above $x = f\cos\theta$, $y = f\sin\theta$, and multiplying by 2 in order that the integral, instead of being taken from 0 to 2π, may be taken from 0 to π, becomes

$$= 2f^s \int_0^\pi \frac{(\sin\theta)^{s-1}\,d\theta}{(f^2+\kappa^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q}}.$$

Write $\cos\frac{1}{2}\theta = \sqrt{x}$; then $\sin\frac{1}{2}\theta = \sqrt{1-x}$, $\sin\theta = 2\sqrt{x(1-x)}$, $d\theta = -x^{-\frac{1}{2}}(1-x)^{-\frac{1}{2}}\,dx$, $\cos\theta = -1+2x$; $\theta = 0$ gives $x = 1$, $\theta = \pi$ gives $x = 0$, and the integral is

$$= 2^{s-1}\int_0^1 \frac{x^{\frac{1}{2}s-1}(1-x)^{\frac{1}{2}s-1}\,dx}{\{(f+\kappa)^2-4f\kappa x\}^{\frac{1}{2}s+q}} = \frac{2^{s-1}}{\{(f+\kappa)^2\}^{\frac{1}{2}s+q}}\int_0^1 \frac{x^{\frac{1}{2}s-1}(1-x)^{\frac{1}{2}s-1}\,dx}{(1-ux)^{\frac{1}{2}s+q}},$$

if for shortness $u = \dfrac{4\kappa f}{(\kappa+f)^2}$, so that obviously $u < 1$.

The integral in x is here an integral belonging to the general form

$$\Pi = \Pi(\alpha,\beta,\gamma,u) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}(1-ux)^{-\gamma}\,dx,$$

viz. we have

$$\text{Ring-integral} = \frac{2^{s-1}f^s}{(f+\kappa)^{s+2q}}\,\Pi(\tfrac{1}{2}s,\tfrac{1}{2}s,\tfrac{1}{2}s+q,u).$$

46–2

[p. 364] **71.** The general function $\Pi(\alpha,\beta,\gamma,u)$ is

$$\Pi(\alpha,\beta,\gamma,u) = \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\alpha+\beta)}\,F(\alpha,\gamma,\alpha+\beta,u),$$

or, what is the same thing,

$$\Pi(\alpha,\beta,\gamma,u) = \frac{\Gamma\gamma}{\Gamma\alpha\,\Gamma(\gamma-\alpha)}\,F(\alpha,\beta,\gamma,u),$$

and consequently transformable by means of various theorems for the transformation of the hypergeometric series, in particular, by the theorems

$$F(\alpha,\beta,\gamma,u) = F(\beta,\alpha,\gamma,u),$$

$$F(\alpha,\beta,\gamma,u) = (1-u)^{\gamma-\alpha-\beta}\,F(\gamma-\alpha,\gamma-\beta,\gamma,u);$$

and if $v = \left(\dfrac{1-\sqrt{1-u}}{1+\sqrt{1-u}}\right)^2$, or, what is the same thing, $u = \dfrac{4\sqrt{v}}{(1+\sqrt{v})^2}$, then

$$F(\alpha,\beta,2\beta,u) = (1-\sqrt{v})^{2\alpha}\,F\big(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v\big).$$

In verification, observe that if $u = 1$ then also $v = 1$, and that with these values, calculating each side by means of the formulae

$$F(\alpha,\beta,\gamma,1) = \frac{\Gamma\gamma\,\Gamma(\gamma-\alpha-\beta)}{\Gamma(\gamma-\alpha)\,\Gamma(\gamma-\beta)},$$

the resulting equation $F(\alpha,\beta,2\beta,1) = 2^{2\alpha}\,F(\alpha,\alpha-\beta+\frac{1}{2},\beta+\frac{1}{2},1)$, becomes

$$\frac{\Gamma 2\beta\,\Gamma(\beta-\alpha)}{\Gamma(2\beta-\alpha)\,\Gamma\beta} = 2^{2\alpha}\,\frac{\Gamma(\beta+\frac{1}{2})\,\Gamma(2\beta-2\alpha)}{\Gamma(2\beta-\alpha)\,\Gamma(\beta-\alpha+\frac{1}{2})},$$

that is,

$$\frac{\Gamma 2\beta}{\Gamma\beta\,\Gamma(\beta+\frac{1}{2})} = 2^{2\alpha}\,\frac{\Gamma(2\beta-2\alpha)}{\Gamma(\beta-\alpha)\,\Gamma(\beta-\alpha+\frac{1}{2})},$$

which is true, in virtue of the relation $\dfrac{\Gamma 2x}{\Gamma x\,\Gamma(x+\frac{1}{2})} = \dfrac{2^{2x-1}}{\Gamma\frac{1}{2}}$.

72. The foregoing formulae, and in particular the formula which I have written

$$F(\alpha,\beta,2\beta,u) = (1-\sqrt{v})^{2\alpha}\,F(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v),$$

are taken from Kummer's Memoir, "Ueber die hypergeometrische Reihe," *Crelle*, t. XV. (1836), viz. the formula in question is, under a slightly different form, his formula (41), p. 76; the formula (43), p. 77, is intended to be equivalent thereto; but there is an error of transcription, $2\alpha-2\beta+1$, in place of $\alpha+\beta-1$, which makes the formula (43) erroneous.

It may be remarked as to the formulae generally that, although very probably $\Pi(\alpha,\beta,\gamma,u)$ may denote a proper function of u, whatever be the values of the indices (α,β,γ), and the various transformation-theorems hold good accordingly (the Γ-function of a negative argument being interpreted in the usual manner by means of the equation $\Gamma x = \dfrac{1}{x}\Gamma(1+x) = \dfrac{1}{x(x+1)}\Gamma(2+x)$, &c.), yet that the function $\Pi(\alpha,\beta,\gamma,u)$,

[p. 365] used as denoting the definite integral $\displaystyle\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}(1-ux)^{-\gamma}\,dx$, has no meaning except in the case where α and β are each of them positive.

In what follows we obtain for the ring-integral and the disk-integral various expressions in terms of Π-functions, which are afterwards transformed into t-integrals with a superior limit ∞ and inferior limit 0, or $= f^2-\kappa^2$; but for values of the variable index, q lying beyond certain limits, the indices α and β, or one of them, of the Π-function will become negative, viz. the integral represented by the Π-function, or, what is the same thing, the t-integral, will cease to have a determinate value, and at the same time, or usually so, the argument or arguments of one or more of the Γ-functions will become negative. It is quite possible that in such cases the results are not without meaning, and that an interpretation for them might be found; but they have not any obvious interpretation, and we must in the first instance consider them as inapplicable.

73. We require further properties of the Π-functions. Starting with the foregoing equation

$$F(\alpha,\beta,2\beta,u) = (1+\sqrt{v})^{2\alpha}\,F\big(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v\big),$$

each side may be expressed in a fourfold form:

$$\begin{aligned}
F(\alpha,\beta,2\beta,u) &= (1+\sqrt{v})^{2\alpha}\,F(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v)\\
&= F(\beta,\alpha,2\beta,u) \;=\; \tfrac{1}{2}(1+\sqrt{v})^{2\alpha-1}\,F(\alpha-\tfrac{1}{2}+1,\,\beta+\tfrac{1}{2},\,v)\\
&= (1-u)^{\beta-\alpha}\,F(2\beta-\alpha,\,\beta,\,2\beta,\,u)\\
&= (1-u)^{\beta-\alpha}\,F(\alpha,\,2\beta-\alpha,\,2\beta,\,u)\\
&= (1+\sqrt{v})^{2\alpha}(1-v)^{2\beta-2\alpha-1}\,F(\beta-\alpha+\tfrac{1}{2},\,2\beta-\alpha,\,\beta+\tfrac{1}{2},\,v)\\
&= (1+\sqrt{v})^{2\alpha}(1-v)^{-\alpha-\beta-\frac{1}{2}}\,F(2\beta-\alpha,\,\beta-\alpha+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v),
\end{aligned}$$

where, instead of $(1+\sqrt{v})^{2\alpha}(1-v)^{2\beta-2\alpha-1}$, it is proper to write $(1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{2\beta-2\alpha-1}$;

And then to each form applying the transformation

$$F(\alpha,\beta,\gamma,u) = \frac{\Gamma\gamma}{\Gamma\alpha\,\Gamma(\gamma-\alpha)}\,\Pi(\alpha,\gamma-\alpha,\beta,u),$$

we have

$$\begin{aligned}
&\frac{\Gamma 2\beta}{\Gamma\alpha\,\Gamma(2\beta-\alpha)}\,\Pi(\alpha,\,2\beta-\alpha,\,\beta,\,u)\\
&= \frac{\Gamma 2\beta}{\Gamma\beta\,\Gamma\beta}\,\Pi(\beta,\beta,\alpha,u)\\
&= (1-u)^{\beta-\alpha}\,\frac{\Gamma 2\beta}{\Gamma(2\beta-\alpha)\,\Gamma\alpha}\,\Pi(2\beta-\alpha,\,\alpha,\,\beta,\,u)
\end{aligned}$$

[p. 366]

$$= (1-u)^{\beta-\alpha}\, 2^{2\alpha-1}\, \frac{\Gamma(\beta+\frac{1}{2})}{\Gamma\frac{1}{2}\,\Gamma(2\beta-\alpha)}\, \Pi(\alpha,\, \alpha-\beta+\tfrac{1}{2},\, 2\beta-\alpha,\, v)$$

$$= (1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{2\beta-2\alpha-1}\, \frac{\Gamma(\beta+\frac{1}{2})}{\Gamma(\beta-\alpha+\frac{1}{2})\,\Gamma\alpha}\, \Pi(\beta-\alpha+\tfrac{1}{2},\, \alpha,\, 2\beta-\alpha,\, v).$$

I select the second of the first four forms; equating it successively to each of the second four forms, and attending to the relation $\dfrac{\Gamma 2\beta}{\Gamma\beta\,\Gamma\beta} = 2^{2\beta-1}\,\dfrac{1}{\Gamma\frac{1}{2}}$, we find

$$\Pi(\beta,\beta,\alpha,u) = (1-\sqrt{v})^{2\alpha}\, 2^{1-2\alpha}\, \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\beta-\alpha+\frac{1}{2})\,\Gamma\beta}\, \Pi(\alpha,\, \beta-\alpha+\tfrac{1}{2},\, \alpha-\beta+\tfrac{1}{2},\, v)$$

$$= (1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{2\beta-2\alpha-1}\, 2^{1-2\alpha}\, \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\beta-\alpha+\frac{1}{2})\,\Gamma\alpha}\, \Pi(\beta-\alpha+\tfrac{1}{2},\, \alpha,\, 2\beta-\alpha,\, v),$$

$$= (1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{-\alpha-\beta+\frac{1}{2}}\, 2^{-1}\, \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(2\beta-\alpha)\,\Gamma\alpha}\, \Pi(2\beta-\alpha,\, \alpha-\beta+\tfrac{1}{2},\, \beta-\alpha+\tfrac{1}{2},\, v).$$

Putting herein $\beta = \frac{1}{2}s$, $\alpha = \frac{1}{2}s + q$, the formulae become

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u) = (1+\sqrt{v})^{-s-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, \Pi(\tfrac{1}{2}s+q,\, \tfrac{1}{2}-q,\, \tfrac{1}{2}+q,\, v) \qquad \text{(I.)}$$

$$= (1+\sqrt{v})^{-s-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\, \Pi(\tfrac{1}{2}+q,\, \tfrac{1}{2}s-q,\, \tfrac{1}{2}s+q,\, v) \qquad \text{(II.)}$$

$$= (1+\sqrt{v})^{-s}(1-\sqrt{v})^{-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\, \Pi(\tfrac{1}{2}-q,\, \tfrac{1}{2}s+q,\, \tfrac{1}{2}s-q,\, v) \qquad \text{(III.)}$$

$$= (1+\sqrt{v})^{-s}(1-\sqrt{v})^{-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\, \Pi(\tfrac{1}{2}s-q,\, \tfrac{1}{2}+q,\, \tfrac{1}{2}-q,\, v) \qquad \text{(IV.)}$$

where observe that on the right-hand side the Π-functions in I. and IV. only differ by the sign of q, and so also the Π-functions in II. and III. only differ by the sign of q. We hence have

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s-q, u) = (1+\sqrt{v})^{-s+2q}\, 2^{q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\, \Pi(\tfrac{1}{2}-q,\, \tfrac{1}{2}s+q,\, \tfrac{1}{2}s-q,\, v);$$

[p. 367] and comparing with (IV.),

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u) = \left(\frac{1+\sqrt{v}}{1-\sqrt{v}}\right)^{2q} \Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s-q, u).$$

74. The foregoing formula,

$$\text{Ring-integral} = \frac{2^{s-1} f^s}{(f+\kappa)^{s+2q}}\, \Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u),$$

where $u = \dfrac{4\kappa f}{(f+\kappa)^2}$, gives, as well in the case of an exterior as an interior point, a convergent series for the integral; but this series proceeds according to the powers of $\dfrac{4\kappa f}{(f+\kappa)^2}$. We may obtain more convenient formulae applying to the cases of an internal and an external point respectively.

75. For an internal point $\kappa < f$, $\sqrt{1-u} = \dfrac{f-\kappa}{f+\kappa}$, and therefore $v = \dfrac{\kappa^2}{f^2}$.

$$\begin{aligned}
\Pi(\tfrac12 s, \tfrac12 s, \tfrac12 s + q, u) &= \left(\frac{f+\kappa}{f}\right)^{s+2q} 2^{-q}\, \frac{\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12 s+q)\,\Gamma(\frac12-q)}\, \Pi\Big(\tfrac12 s+q,\ \tfrac12-q,\ \tfrac12+q,\ \tfrac{\kappa^2}{f^2}\Big)\\
&= \left(\frac{f+\kappa}{f}\right)^{s+2q} 2^{-q}\, \frac{\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12+q)\,\Gamma(\frac12 s-q)}\, \Pi\Big(\tfrac12+q,\ \tfrac12 s-q,\ \tfrac12 s+q,\ \tfrac{\kappa^2}{f^2}\Big)\\
&= \left(\frac{f+\kappa}{f}\right)^{s} \left(\frac{f-\kappa}{f}\right)^{-2q} 2^{-q}\, \frac{\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12-q)\,\Gamma(\frac12 s+q)}\, \Pi\Big(\tfrac12-q,\ \tfrac12 s+q,\ \tfrac12 s-q,\ \tfrac{\kappa^2}{f^2}\Big)\\
&= \left(\frac{f+\kappa}{f}\right)^{s} \left(\frac{f-\kappa}{f}\right)^{-2q} 2^{-q}\, \frac{\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12 s-q)\,\Gamma(\frac12+q)}\, \Pi\Big(\tfrac12 s-q,\ \tfrac12+q,\ \tfrac12-q,\ \tfrac{\kappa^2}{f^2}\Big);
\end{aligned}$$

where the Π-functions on the right-hand side are respectively

$$\int_0^1 \frac{x^{\frac12 s+q-1}(1-x)^{-q-\frac12}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac12+q}} = \int_0^\infty \frac{t^{\frac12 s+q-1}\,dt}{(t+f^2-\kappa^2)^{q+\frac12}(t+f^2)^{\frac12 s+q-\frac12}}\cdot f^{s-1},$$

$$\int_0^1 \frac{x^{q-\frac12}(1-x)^{\frac12 s-q-1}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac12 s+q}} = \int_0^\infty \frac{t^{q-\frac12}(t+f^2-\kappa^2)^{\frac12 s-q-1}\,dt}{(t+f^2)^{\frac12 s+q}}\cdot f^{-1},$$

$$\int_0^1 \frac{x^{-q-\frac12}(1-x)^{\frac12 s+q-1}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac12 s-q}} = \int_0^\infty \frac{t^{-q-\frac12}(t+f^2-\kappa^2)^{\frac12 s+q-1}\,dt}{(t+f^2)^{\frac12 s-q}}\cdot f^{-1},$$

$$\int_0^1 \frac{x^{\frac12 s-q-1}(1-x)^{q-\frac12}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac12-q}} = \int_0^\infty \frac{t^{\frac12 s-q-1}(t+f^2-\kappa^2)^{q-\frac12}\,dt}{(t+f^2)^{\frac12+q}}\cdot f^{-1},$$

the t-forms being obtained by means of the transformation $x = \dfrac{t}{t+f^2-\kappa^2}$; viz. this gives

$$1-x = \frac{f^2-\kappa^2}{t+f^2-\kappa^2},\quad 1-\tfrac{\kappa^2}{f^2}\,x = \frac{t+f^2}{t+f^2-\kappa^2}\cdot\frac{1}{f^2},\quad dx = \frac{(f^2-\kappa^2)\,dt}{(t+f^2-\kappa^2)^2},$$

whence the results just written down.